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Trajectory Prediction for Surface Drifters with Calibrated Search Regions

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Abstract

Predicting where a drifting object will be within hours to days is essential for maritime search and rescue. Using hourly trajectories of 1352 GPS-tracked Global Drifter Program drifters in the Northwest Pacific and North Atlantic (2015–2022), we develop and retrospectively evaluate a probabilistic drift forecasting framework based on trajectory history. The framework combines an Ornstein–Uhlenbeck (OU; Langevin) prior with a long short-term memory (LSTM) network that predicts a bivariate Gaussian correction, and employs rolling Mondrian conformal calibration to construct search regions for 6–72 h forecasts. In both basins, the hybrid reduces the mean position error of damped-velocity extrapolation by 32% at 24 h and 18–19% at 72 h. Alternative neural architectures achieve similar accuracy, with differences within 4%. The physical prior provides little additional accuracy when training data are abundant but improves robustness when only one or two drifters are available. Under interannual distribution shifts, rolling Mondrian calibration restores overall coverage to approximately 90% and improves consistency across latitude–drogue groups. At comparable coverage, the hybrid produces search regions 47–61% smaller at 24 h and 29–46% smaller at 72 h than those of the OU prior.

Keywords: surface drifters; drift forecasting; Ornstein–Uhlenbeck process; LSTM; conformal prediction; maritime search and rescue

1. Introduction

Knowing where a drifting object will be in the next hours to days is central to maritime search and rescue (SAR), oil-spill response and the tracking of marine debris [1–3]. Operational systems usually advect virtual particles with forecast ocean currents, winds and waves, adding object-specific leeway and stochastic perturbations to obtain a probabilistic search area [1,4,5]. Their skill is limited by the quality of the forcing fields, which do not resolve the submesoscale and inertial motions that dominate drift over the first day.

A complementary source of information is the trajectory of the object itself, or of nearby drifting buoys. The NOAA Global Drifter Program (GDP) has accumulated decades of quality-controlled drifter trajectories [6–8], and these data have motivated statistical and machine-learning approaches that learn drift directly from observed tracks. Stochastic Lagrangian models describe drifter velocities as Ornstein–Uhlenbeck (OU) or higher-order processes [9–12]; more recently, recurrent, convolutional and attention-based networks have been trained to predict drifter or debris trajectories [13–18].

Two requirements of SAR shape how such models should be built and evaluated. The first is that a forecast can only use information available at the time it is issued. In a real search the object—a person in the water, a life raft or a marker buoy deployed by the rescue unit—provides only the GPS fixes received so far, and no product velocity is

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available. The velocities distributed with drifter products, in contrast, are estimated after the fact from positions on both sides of each time: the GDP 6-hourly positions are obtained by kriging that uses fixes before and after each estimation time, and the velocities are central differences of these positions over 12 h [6,7,19,20]; the hourly product estimates positions and velocities with locally weighted scatterplot smoothing (LOWESS), a local polynomial fit centred on each time [20,21]. These products are designed for oceanographic analysis, where using later positions improves the estimates; for forecasting, however, they contain information about the motion after the issue time, which in machine learning is known as information leakage [22,23]. For the same reason, overlapping windows from one trajectory should not be split between training and test sets. The second requirement concerns uncertainty. A point forecast is of limited use in SAR, where the quantity of interest is a search region that contains the object with a known probability and is as small as possible. Probabilistic deep learning [24–26] provides such regions, but their coverage must hold across years, regions and drifter types.

This study addresses both requirements for 6–72 h drift forecasts from trajectory history alone, in two ocean basins. Its contributions are:

1. We define a forecasting protocol for GPS-tracked drifters that uses backward-difference velocities and separates development and test periods by forecast issue time, with a spatial hold-out for transfer tests, and quantify why the choice of velocity matters: the centred velocity of the 6-hourly product, emulated from hourly data, would reduce the 24-h error of a constant-velocity forecast by 30–36%, whereas the hourly product velocity changes it by only 1%. A sensitivity experiment in which future positions are added to the input shows how strongly the forecasts respond to such information; residual look-ahead from the smoothing of the hourly positions is not fully excluded.
2. We combine an OU (Langevin) prior, which gives an analytic mean and variance of the future displacement, with a long short-term memory (LSTM) network [27] that predicts a bivariate Gaussian residual on top of it. The gain over the best deterministic baseline is nearly identical in the Northwest Pacific and the North Atlantic and insensitive to the architecture—a multilayer perceptron and a sequence-to-sequence model with attention perform within 4%, gradient-boosted trees 5–7% worse—and the prior turns out to matter only when very few drifters are available, where it makes the failures of a purely data-driven network much less frequent.
3. We evaluate standard split conformal calibration [28,29] under interannual shift and find that it loses its nominal coverage, whereas a rolling, group-conditional (Mondrian) variant restores the overall empirical coverage to 0.89–0.90 and brings most latitude–drogue groups close to 90%, although sparse groups under spatial transfer can remain under-covered (down to 0.79). At comparable coverage the hybrid model shrinks the search region by 29–46% at 72 h and 47–61% at 24 h relative to the OU prior.

Section 2 describes the data, the choice of input velocities and the forecasting protocol; Section 3 the models and the calibration; Section 4 the results; and Sections 5 and 6 discuss limitations and conclude.

2. Data and Forecasting Protocol

2.1. Hourly Drifter Trajectories

We use version 2.01 of the GDP hourly dataset [8,20], which provides positions and velocities at the top of each hour together with their uncertainties and the date of drogue loss, from October 1987 to 31 October 2022. Trajectories from January 2015 onwards were extracted for two boxes (Figure 1): the Northwest Pacific (NWP; 20–45°N, 120–180°E), which contains the Kuroshio, its extension and the western subtropical gyre, and the North

Atlantic (NA; 25–50°N, 80–10°W), which contains the Gulf Stream, the North Atlantic Current and the eastern subtropical gyre. Each box was split into a western and an eastern sub-region (at 150°E and 45°W) for transfer experiments.

Only GPS-tracked drifters were kept. For these, the hourly positions result from a LOWESS fit of the quality-controlled fixes with a tricube kernel whose half-width spans two fixes on either side of each fix, followed by linear interpolation to hourly times, so that the hourly position at a given time depends on fixes received shortly after that time. In the GPS data analysed by Elipot et al. [20], the non-zero differences between the quality-controlled fixes and the smoothed positions had a median of 30 m (90th percentile 77 m); these values were not recomputed for the present data. The positioning system was identified from the corrected metadata of release 2.01.1, because the distribution of release 2.01 flags every drifter as GPS-tracked. This leaves 508 drifters (2.8 million hourly positions) in the NWP and 844 drifters (8.5 million) in the NA. A drifter is considered drogued before its drogue-loss date and undrogued afterwards [30]; 30% (NWP) and 21% (NA) of the test windows are drogued and 9% and 5% have an unknown drogue state.

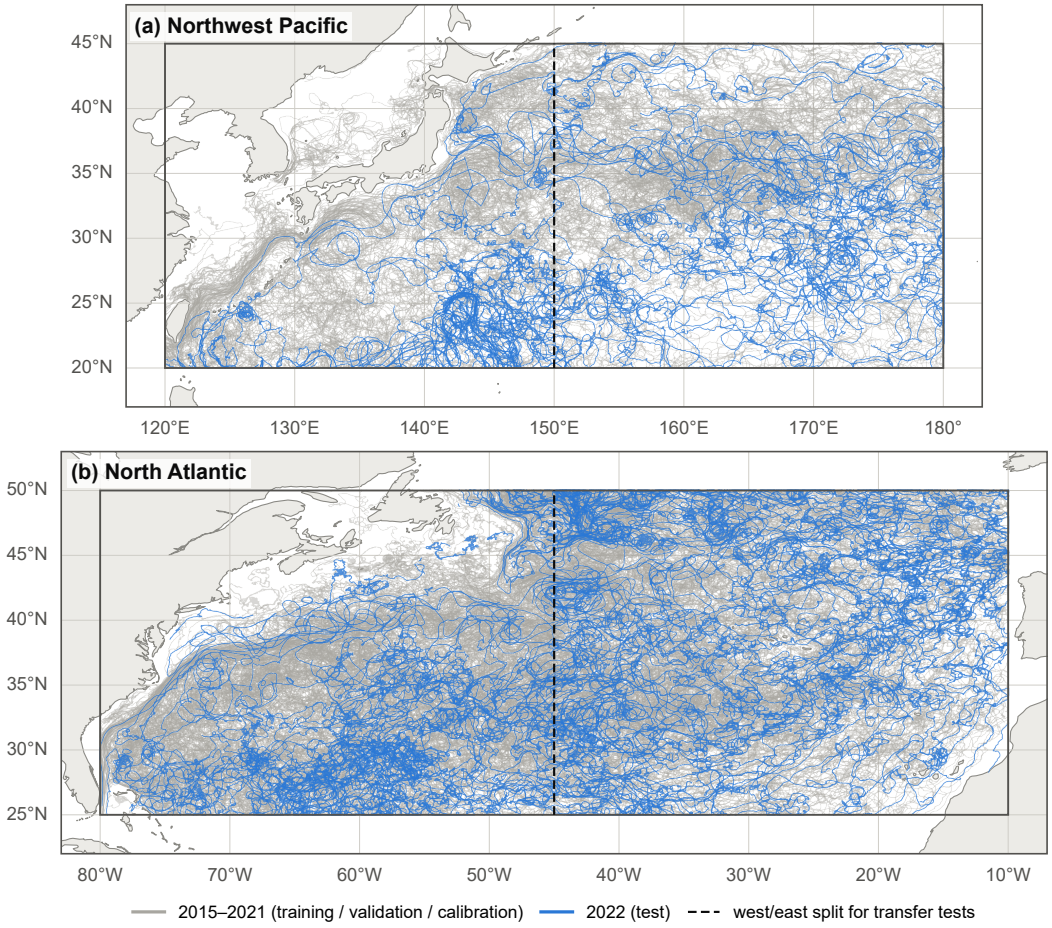


Figure 1. Hourly GPS drifter trajectories in the two study boxes: (**a**) Northwest Pacific, 20–45°N, 120–180°E; (**b**) North Atlantic, 25–50°N, 80–10°W. Grey: 2015–2021, used for training, early stopping and calibration; blue: 2022 test period. Dashed lines separate the western and eastern sub-regions used for the spatial transfer experiments.

2.2. Forecast Windows

Each trajectory was divided into segments of consecutive hourly estimates. A forecast is issued every 6 h at time t_0 if the segment contains the previous 48 h and the following 72 h; windows in which any position relies on a fix gap longer than 6 h were discarded

(30,864 windows in the NWP and 55,539 in the NA). Positions are expressed in a local tangent plane centred on the drifter position at t_0 (east and north, in km). The model input is the history at 3-h spacing (17 time steps) of position, velocity obtained by backward differences of positions, and speed, together with the season (sin and cos of the day of year) and the drogue state (drogued, undrogued or unknown, one-hot encoded). The target is the displacement $\mathbf{y}(\tau)$ at horizons $\tau = 6, 12, 24, 48$ and 72 h. The product velocities (v_e, v_n) and the absolute latitude and longitude are not used as inputs (the latter are discussed in Section 4.4).

In each basin, windows issued in 2015–2021 were split by drifter into training (70%), early-stopping validation (15%) and conformal calibration (15%) sets; all windows issued in 2022 form the test set (Table 1). Windows were assigned to the two periods by their issue time, so drifters alive across the year boundary contribute to both periods, and the targets of their last 2021 windows (at most 72 h into 2022) overlap in time with their first test windows.

Table 1. GPS drifters and forecast windows (number of drifters in parentheses). Windows are issued every 6 h and require 48 h of history and 72 h of future positions.

Basin	Drifters	Hourly positions	Training	Validation	Calibration	Test (2022)
Northwest Pacific	508	2,825,297	230,947 (292)	49,996 (62)	63,828 (63)	56,539 (121)
North Atlantic	844	8,507,955	781,928 (518)	160,477 (111)	168,673 (111)	192,841 (265)

2.3. Product Velocities and Information at Issue Time

To quantify how much the choice of input velocity matters, we compared constant-velocity forecasts $\hat{\mathbf{y}}(\tau) = \mathbf{u}\,\tau$ that differ only in how $\mathbf{u}$ is estimated at t_0: (i) a past-only backward difference over 3 h; (ii) the hourly product velocity; (iii) a central difference over ± 1 h, which uses the position at $t_0 + 1$ h; and (iv) a central difference over ± 6 h, which emulates the velocity of the 6-hourly product [7,19] from the hourly positions. Across all hourly estimates, the product velocity agrees best with central differences (correlation 0.98 for both components in both basins; root-mean-square (RMS) difference 0.090 and 0.068 $\mathrm{m\,s^{-1}}$ in the NWP and NA, compared with 0.116 and 0.091 $\mathrm{m\,s^{-1}}$ for backward differences), consistent with its symmetric LOWESS estimation.

3. Models and Calibration

Figure 2 gives an overview of the forecasting framework.

3.1. Physical Baselines and the OU Prior

Persistence predicts zero displacement. *Constant velocity* extrapolates the past-only velocity $\mathbf{u}_0$ (3-h backward difference). *Damped velocity* assumes that the velocity anomaly decays exponentially, $\hat{\mathbf{y}}(\tau) = \mathbf{u}_0\, T_d\, (1 - e^{-\tau/T_d})$, with T_d chosen on the training windows ($T_d = 36$ h in both basins).

The *OU prior* treats the drifter velocity as a background velocity $\alpha\bar{\mathbf{u}}$ plus an OU anomaly with decorrelation time T and stationary variance σ^2 per component [9,11], where $\bar{\mathbf{u}}$ is the least-squares mean velocity over the 48-h history. Conditioned on the current velocity $\mathbf{u}_0$, the displacement is Gaussian with mean and isotropic variance

$$\boldsymbol{\mu}_{\mathrm{OU}}(\tau) = \alpha\bar{\mathbf{u}}\,\tau + (\mathbf{u}_0 - \alpha\bar{\mathbf{u}})\,T\left(1 - e^{-\tau/T}\right), \tag{1}$$

$$\sigma^2_{\mathrm{OU}}(\tau) = \sigma^2 T^2 \left[\frac{2\tau}{T} - 3 + 4e^{-\tau/T} - e^{-2\tau/T}\right]. \tag{2}$$

The parameters $\alpha \in \{0, 0.25, \ldots, 1\}$ and T were selected on the training windows by the mean separation over all horizons, and σ^2 was then obtained by maximum likelihood. The fitted values are $\alpha = 0.5$, $T = 12$ h and $\sigma = 0.31$ m s^{-1} in the NWP, and $\alpha = 0.5$, $T = 24$ h and $\sigma = 0.27$ m s^{-1} in the NA. For short horizons Equation (2) grows as τ^3 (the velocity is known at t_0) and for long horizons as $2\sigma^2 T\tau$, the classical diffusive limit [11].

3.2. Probabilistic LSTM with an OU Prior

The network encodes the 17-step history with a two-layer LSTM (64 units, dropout 0.1); its last hidden state is concatenated with the static features and passed through a multilayer perceptron (128 units) that outputs, for each horizon, five numbers $(\Delta\mu_x, \Delta\mu_y, a, b, c)$. These define a bivariate Gaussian

$$\boldsymbol{\mu} = \boldsymbol{\mu}_{\mathrm{OU}} + \sigma_{\mathrm{OU}}\,\Delta\boldsymbol{\mu}, \qquad \sigma_x = \sigma_{\mathrm{OU}}\,e^{a}, \quad \sigma_y = \sigma_{\mathrm{OU}}\,e^{b}, \quad \rho = 0.95\tanh c, \tag{3}$$

so that the network learns corrections to the prior in units of the prior standard deviation. The output layer is initialised to zero; training therefore starts exactly at the OU prior. The *LSTM direct* variant replaces $\boldsymbol{\mu}_{\mathrm{OU}}$ by zero and σ_{OU} by the root-mean-square training displacement at each horizon. Both are trained by minimising the bivariate Gaussian negative log-likelihood (NLL) averaged over horizons [24], with Adam [31] (learning rate 2×10^{-3}, cosine decay, batch size 512) for at most 30 epochs with early stopping on the validation NLL (patience 4), in PyTorch [32]. Each model is an ensemble of three networks with different seeds, combined by moment matching of the Gaussian mixture [25].

3.3. Alternative Models

To test whether the results depend on the network architecture, three alternatives were trained with the same inputs, outputs, split and calibration: (i) a multilayer perceptron (MLP) on the flattened 48-h history and the static features, with two hidden layers of 256 units; (ii) a sequence-to-sequence model with attention (Seq2Seq-Att), a common design in trajectory forecasting, in which an LSTM encoder (two layers, 64 units) is followed by an autoregressive LSTM decoder over the five horizons with Luong (general) attention on the encoder states; and (iii) gradient-boosted regression trees (scikit-learn HistGradientBoosting, at most 500 iterations with early stopping on an internal 10% hold-out, one model per horizon and component, trained on up to 300,000 training windows, i.e., all NWP training windows and a random subset in the NA) on the same flattened features, which provide point forecasts only. Each alternative is trained both directly and relative to the OU prior: the networks use the parameterisation of Equation (3) with a zero-initialised output layer, and the trees predict the residual from $\boldsymbol{\mu}_{\mathrm{OU}}$ scaled by σ_{OU}. The networks are ensembles of three seeds, trained as described above.

3.4. Conformal Calibration of Search Regions

For a forecast with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, the nonconformity score is the squared Mahalanobis distance $d^2 = (\mathbf{y} - \boldsymbol{\mu})^{\top}\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$, and the $(1 - \alpha_c)$ search region is the ellipse $\{\mathbf{y} : d^2 \le q\}$ with area $\pi q\,\sigma_x\sigma_y\sqrt{1-\rho^2}$. Without calibration $q = \chi^2_{2,0.9} = 4.605$. *Static* split conformal prediction [28,29] sets q per horizon to the $\lceil (n+1)(1-\alpha_c)\rceil/n$ empirical quantile of d^2 on the n calibration windows. Because exchangeability between calibration and test years cannot be assumed [33–35], we also use a *rolling* variant: for a forecast issued on day D, q is the same quantile computed from forecasts in the target stream whose verification time $(t_0 + \tau)$ fell within the 30 days before the start of day D (UTC), i.e., forecasts that an operator could have verified (verification times are taken as $t_0 + \tau$, neglecting transmission and processing delays and the few hours needed by the position smoothing). If fewer than 100 such forecasts are available, the window is widened

to 90 days, and if there are still fewer than 100, the static threshold is used; in particular, the static threshold applies at the start of the test period, before any forecast has been verified. The *Mondrian* variant [28] applies the same rule separately within six groups (three latitude bands $\times$ drogued/undrogued; bands 20–30, 30–38, 38–45°N in the NWP and 25–33, 33–41, 41–50°N in the NA), falling back to the marginal rolling threshold instead of the static one; windows with unknown drogue state are assigned to the undrogued group of their latitude band. We use $\alpha_c = 0.1$.

3.5. Experiments and Metrics

The main experiment trains all models on the training set of each basin and evaluates them on its 2022 test set; the calibration results and the alternative models use the same split. Four additional experiments use the same windows: (i) a location ablation, adding the latitude, longitude and Coriolis parameter at t_0 to the static features; (ii) spatial transfer, training on windows from one sub-region and testing on 2022 windows from the other, with a model trained in the target sub-region as reference; (iii) an oracle sensitivity experiment in which future positions are added to the input (Section 4.4); and (iv) a very-small-data experiment in which the OU prior and both networks are fitted on 1, 2, 4 or 8 drifters drawn at random from those with at least 40 pre-2022 windows (five draws per basin), with early stopping on the last 20% (in time) of the pooled windows of the selected drifters (at most 100 epochs, patience 10, batch size 128); training and early-stopping windows adjacent to this split share up to 120 h of positions. Experiments (i)–(iii) use an 80/20 drifter split of 2015–2021 into training and early-stopping sets and two seeds per network. The calibration results under transfer (Section 4.6) come from separate models trained on the source sub-region with the same 70/15/15 drifter split as the main experiment; their static threshold is computed on the 15% calibration drifters of the source sub-region, which are disjoint from the training and early-stopping drifters, and their rolling thresholds use verified 2022 forecasts from the target sub-region. Accuracy is measured by the mean and median separation between the predicted mean and the observed position; probabilistic quality by the NLL, the empirical coverage of the 90% region, overall and per group, and the median area of the region.

4. Results

4.1. Effect of the Input Velocity

Figure 3a compares constant-velocity forecasts on the 2022 test windows. The product velocity and the $\pm$1-h central difference change the 24-h error by 1% and 4%, respectively, relative to the past-only estimate. The $\pm$6-h central difference, which emulates the velocity of the 6-hourly GDP product, reduces the mean 24-h separation from 17.5 to 12.3 km (30%) in the NWP and from 13.1 to 8.4 km (36%) in the NA, and the 72-h separation by 18% and 23%. This gain reflects both the use of positions up to 6 h after the forecast time, information that does not exist in a real search, and the longer (12-h) differencing span, so it indicates the size of the effect rather than isolating it; it is of the same order as the gain of the LSTM reported below. All models in this study therefore use velocities computed from past positions, so that their differences reflect the methods rather than the inputs.

4.2. Deterministic Accuracy

Table 2 and Figure 4 summarise the test results. Damping the velocity anomaly is the most effective physical refinement: it reduces the 72-h error of the constant-velocity forecast by 23% in the NWP and 26% in the NA. The OU prior, which adds half of the 48-h mean velocity as a persistent background, improves on damped velocity by a further 3–6%. Both LSTM variants reduce the error substantially further, to 10.3 km (NWP) and 7.5 km (NA) at

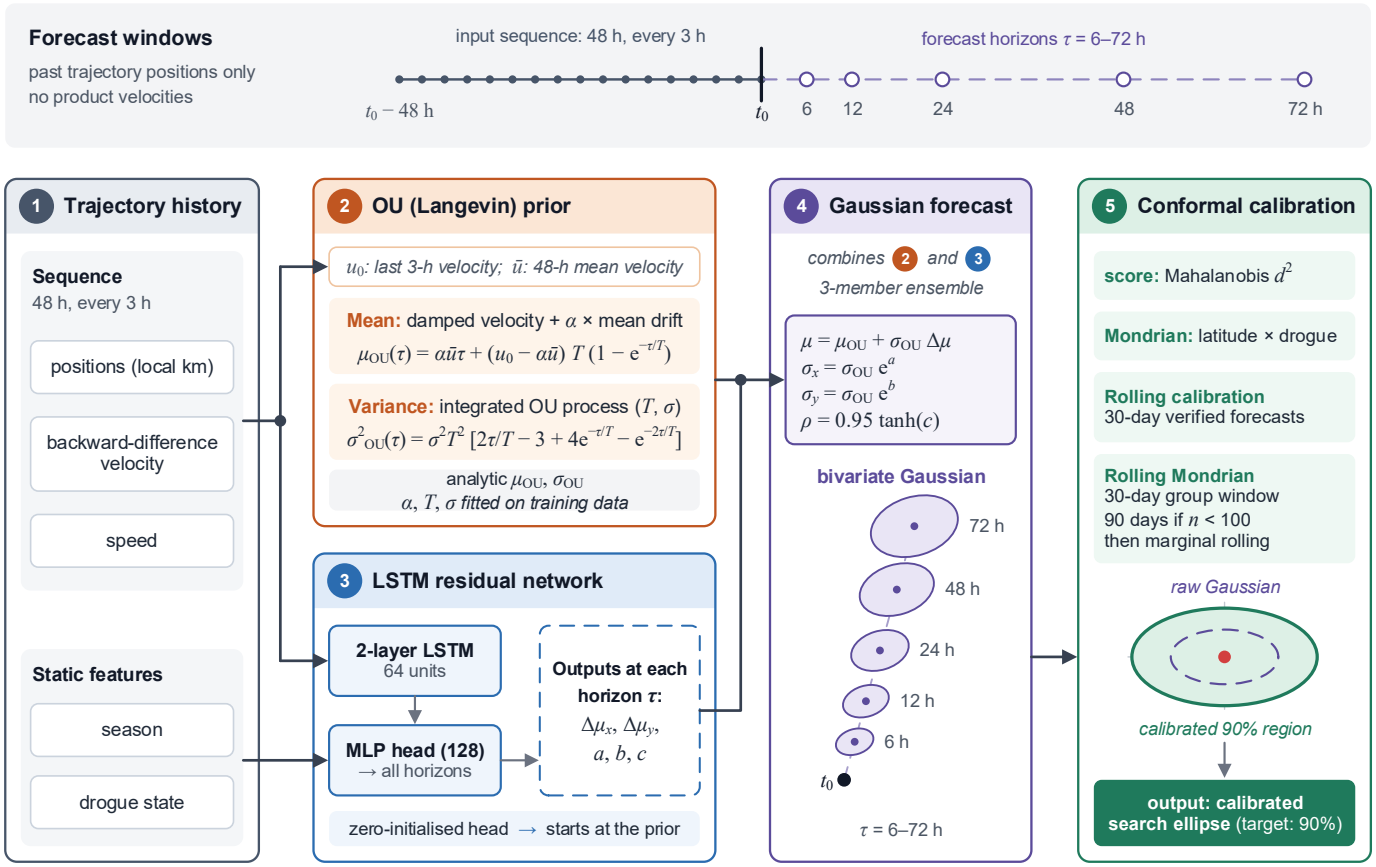


Figure 2. Overview of the probabilistic drift forecasting framework. (1) Trajectory-history inputs are used by (2) the Ornstein–Uhlenbeck (OU) prior to estimate the displacement mean and variance (Equations (1) and (2)), and (3) an LSTM to predict Gaussian residual corrections. Static features enter the prediction head. (4) The two components are combined into bivariate Gaussian forecasts for 6–72 h (Equation (3)) using a three-member ensemble. (5) Rolling conformal calibration uses forecasts verified in the previous 30 days; group-conditional Mondrian calibration applies the same rule within each latitude–drogue group, widening the window to 90 days and then falling back to the marginal rolling threshold when fewer than 100 verified forecasts are available, producing search ellipses targeting 90% coverage.

24 h and 40.3 and 28.8 km at 72 h, i.e., 32% below damped velocity at 24 h and 18–19% at 72 h in both basins. The improvement is largest for drogued drifters (38–40% at 24 h) and smallest in the bands containing the Kuroshio Extension (24%) and the Gulf Stream and North Atlantic Current (28%), where the flow is most energetic (Figure 4c,d). With the full training sets the direct and the prior-based LSTMs are indistinguishable (differences below 0.1 km).

4.3. Comparison with Alternative Models

Table 3 compares the LSTM with the alternative models, evaluated under the same forecasting protocol and with the same input features (training subsets in Section 3.3). All networks reach nearly the same accuracy: in each basin the 24-h errors of the six network variants lie within 4% of each other (NWP: 10.30–10.68 km; NA: 7.53–7.65 km) and the 72-h errors within 2% (40.09–40.87 and 28.71–28.90 km). After rolling Mondrian calibration every network attains 0.90 coverage, with similar 72-h search regions (15.3–17.2 and 6.8–6.9 $\times$ 10^3 km^2). Gradient-boosted trees on the same features are 5–7% less accurate at 24 h and 4–5% at 72 h. With the full training sets the OU prior again leaves the accuracy essentially unchanged, except for the MLP in the NWP (10.68 vs. 10.42 km at 24 h). The skill

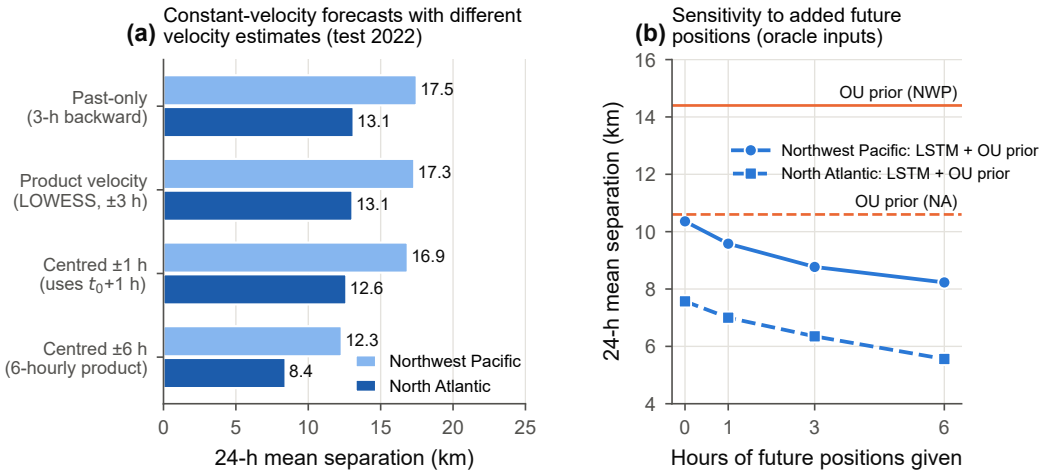


Figure 3. (**a**) Mean 24-h separation of constant-velocity forecasts on the 2022 test windows when the velocity at t_0 is estimated from past positions only, taken from the hourly product or computed by central differences over ± 1 h and ± 6 h (the latter emulating the 6-hourly product). (**b**) Sensitivity to added future information: 24-h error of the LSTM with OU prior when the hourly positions of the first 1, 3 or 6 h after t_0 are added to its input (models trained on the 80/20 split used for experiments (i)–(iii)); horizontal lines show the OU prior.

Table 2. Results on the 2022 test windows. Mean separation at each horizon and median separation at 72 h (Med.; km); NLL averaged over horizons and windows (positions in km); coverage of the 90% region at 72 h before (Raw) and after (Calib.) rolling Mondrian calibration, and median area of the calibrated region (10^3 km^2). Bold: best value within each basin.

Model	Separation (km)						NLL	90% region at 72 h		
	6 h	12 h	24 h	48 h	72 h	Med.		Raw	Calib.	Area
Northwest Pacific										
Persistence	6.82	12.79	23.78	45.15	64.45	55.17	–	–	–	–
Constant velocity	3.94	9.07	17.49	39.82	64.74	54.99	–	–	–	–
Damped velocity	3.80	8.39	15.09	32.21	49.88	42.67	–	–	–	–
OU prior	3.68	7.83	14.13	31.01	48.30	39.29	7.93	0.87	0.90	27.8
LSTM direct	2.32	5.11	10.38	**24.56**	**40.21**	32.81	7.01	0.88	0.90	15.3
LSTM + OU prior	**2.24**	**4.98**	**10.30**	24.58	40.31	**32.69**	7.02	0.88	0.90	16.5
North Atlantic										
Persistence	4.82	8.83	16.47	31.02	44.12	36.86	–	–	–	–
Constant velocity	3.22	6.90	13.13	29.79	47.94	39.46	–	–	–	–
Damped velocity	3.10	6.33	11.13	23.23	35.25	29.17	–	–	–	–
OU prior	3.06	6.14	10.60	22.02	33.56	26.23	7.50	0.96	0.90	12.5
LSTM direct	**1.65**	**3.58**	**7.54**	**17.82**	**28.71**	**22.31**	6.27	0.91	0.90	6.8
LSTM + OU prior	1.65	3.58	7.54	17.84	28.77	22.32	6.27	0.91	0.90	6.8

therefore appears to reflect the information contained in the trajectory and the evaluation protocol rather than a particular architecture, and the OU prior and the conformal step can be combined with any of these networks.

4.4. Is the Gain Due to Position Smoothing or Geography?

Two alternative explanations for the large LSTM gain were tested. First, the hourly positions are themselves LOWESS estimates whose window extends slightly beyond each time, so the inputs may carry some information about the motion after t_0. To gauge the sensitivity of the forecasts to such information, we added the hourly positions after t_0 to the input (Figure 3b; models trained on the 80/20 split used for experiments (i)–(iii)). Adding

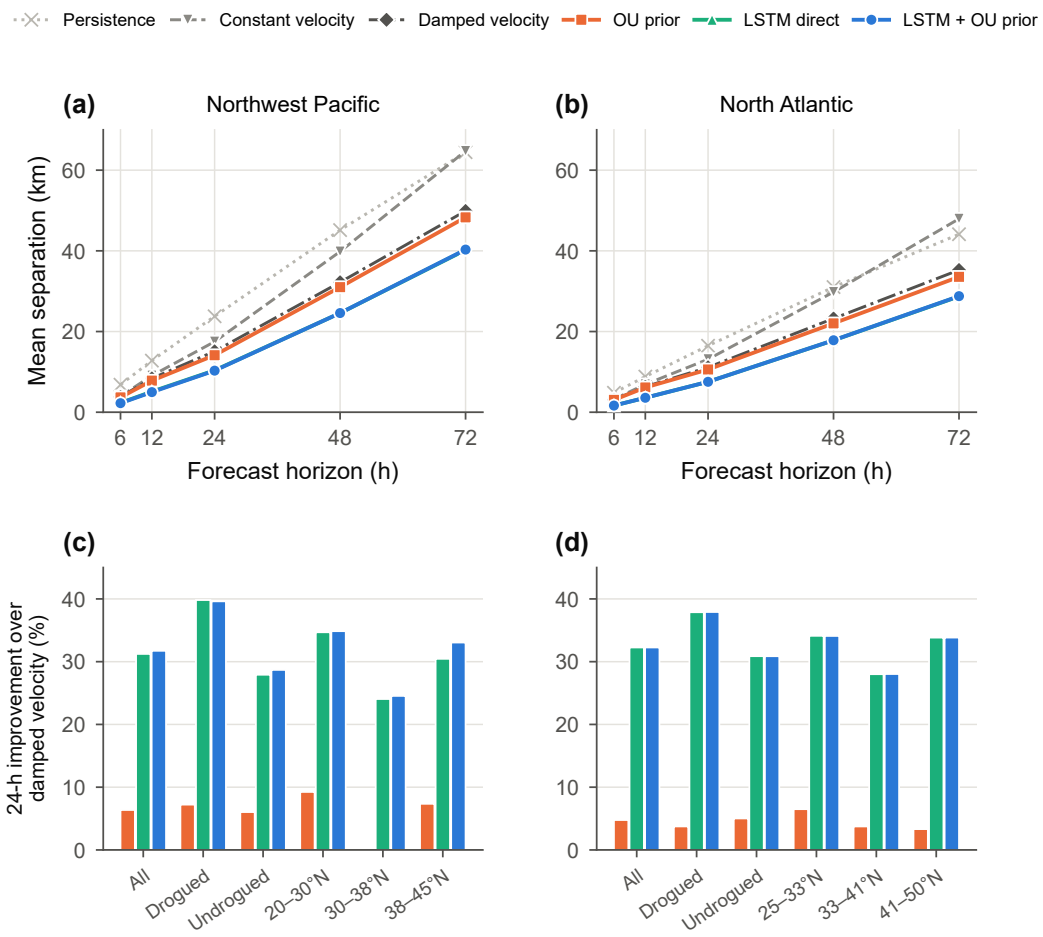


Figure 4. (**a**,**b**) Mean separation versus forecast horizon on the 2022 test windows of the two basins. (**c**,**d**) Reduction of the mean 24-h separation relative to damped velocity, overall, by drogue state and by latitude band.

the position at $t_0 + 1$ h reduces the 24-h error by 0.78 km (NWP) and 0.57 km (NA), and adding the positions up to $t_0 + 3$ h reduces it by 1.59 and 1.22 km, compared with gains of the LSTM over the OU prior of 4.04 and 3.03 km in the same setting. Because the inputs may already contain part of this information, these increments do not bound the effect of the smoothing. The short forward extent of the LOWESS window of the GPS positions (about 2–3 h, with rapidly decreasing weights; [20]) suggests that the residual look-ahead is limited, but removing it would require positions reconstructed from the fixes received before t_0. The physical baselines use the same smoothed positions, although the network may exploit residual look-ahead more effectively.

Second, the network might learn a regional current climatology. Adding latitude, longitude and the Coriolis parameter changes the in-basin 24-h error by at most 0.13 km (NWP: 10.32 vs. 10.19 km; NA: 7.57 vs. 7.56 km for the prior-based network), so the gain comes from the trajectory history rather than from geography, consistent with the better performance of an LSTM trained on relative trajectory intervals rather than absolute trajectory positions [16]. Under spatial transfer the location inputs are unhelpful or harmful: in the NA they raise the west-to-east 24-h error from 6.91 to 7.29–7.87 km (Figure 5c), and in the NWP they lower the raw 72-h coverage of the west-to-east transfer from 0.92 to 0.71. Without location inputs, transfer within the NA costs only 2–3% relative to models trained in the target sub-region. All other experiments therefore exclude location inputs.

Table 3. Comparison with alternative models on the 2022 test windows: mean separation at 24 and 72 h and median separation at 72 h (km), NLL averaged over horizons and windows (positions in km), coverage of the 90% region at 72 h after rolling Mondrian calibration and its median area (10^3 km^2). Gradient boosting provides point forecasts only. Bold: lowest mean separation within each basin.

Model	Variant	Separation (km)			NLL	90% region at 72 h	
		24 h	72 h	Med. 72 h		Coverage	Area
Northwest Pacific							
LSTM	direct	10.38	40.21	32.81	7.01	0.90	15.3
LSTM	+ OU prior	**10.30**	40.31	32.69	7.02	0.90	16.5
MLP	direct	10.68	40.87	33.45	7.07	0.90	16.1
MLP	+ OU prior	10.42	40.57	32.97	7.01	0.90	15.7
Seq2Seq-Att	direct	10.35	**40.09**	32.56	7.05	0.90	16.5
Seq2Seq-Att	+ OU prior	10.31	40.35	32.63	7.07	0.90	17.2
Gradient boosting	direct	10.95	42.13	34.39	–	–	–
Gradient boosting	+ OU prior	10.95	42.04	34.38	–	–	–
North Atlantic							
LSTM	direct	7.54	**28.71**	22.31	6.27	0.90	6.8
LSTM	+ OU prior	7.54	28.77	22.32	6.27	0.90	6.8
MLP	direct	7.65	28.90	22.45	6.32	0.90	6.8
MLP	+ OU prior	7.59	28.86	22.35	6.30	0.90	6.9
Seq2Seq-Att	direct	**7.53**	28.73	22.28	6.27	0.90	6.9
Seq2Seq-Att	+ OU prior	7.54	28.79	22.33	6.27	0.90	6.9
Gradient boosting	direct	8.09	30.01	23.42	–	–	–
Gradient boosting	+ OU prior	8.09	30.02	23.45	–	–	–

4.5. When Does the Physical Prior Help?

With the full training sets the prior adds no accuracy, and the more relevant question is whether it helps when data are scarce—the situation of a SAR operator in a region with few recent drifters. Figure 5a,b shows the 24-h error of networks fitted on one to eight randomly drawn drifters, relative to an OU prior fitted on the same drifters. With a single drifter (median over the five draws per basin: 335 training windows in the NWP and 288 in the NA) the direct LSTM is more than 10% worse than the prior in 8 of the 10 draws, with errors up to 6.5 times that of the prior, whereas the prior-based network is worse in 2 of 10 draws. With two drifters (medians of 1184 and 1751 training windows; drifters are drawn independently for each training-set size, and their record lengths differ widely) the direct network still fails in 6 of 10 draws and the prior-based network in none. With four drifters failures become rare (one draw of the direct network in the NWP), and with eight drifters both networks beat the prior, the direct network slightly more (median ratio 0.81–0.86 against 0.84–0.89). Because the output layer starts at zero and its corrections are scaled by σ_{OU}, a prior-based network starts at the physical solution and, with little data, tends to stay close to it; the corrections are not bounded, however, so this is not a guarantee. The price is a slightly slower departure from the prior when moderately more data are available.

The prior also stabilised training in some transfer settings: in the NWP east-to-west reference, one of two direct networks failed (24-h errors differing by 3.5 km between seeds) while the prior-based networks agreed within 0.07 km, and with location inputs in the NA the seed spread of the direct network was 1.19 km against 0.16 km. The effect is not systematic, however; in the NWP west-to-east transfer the prior-based networks varied more between seeds (Figure 5c).

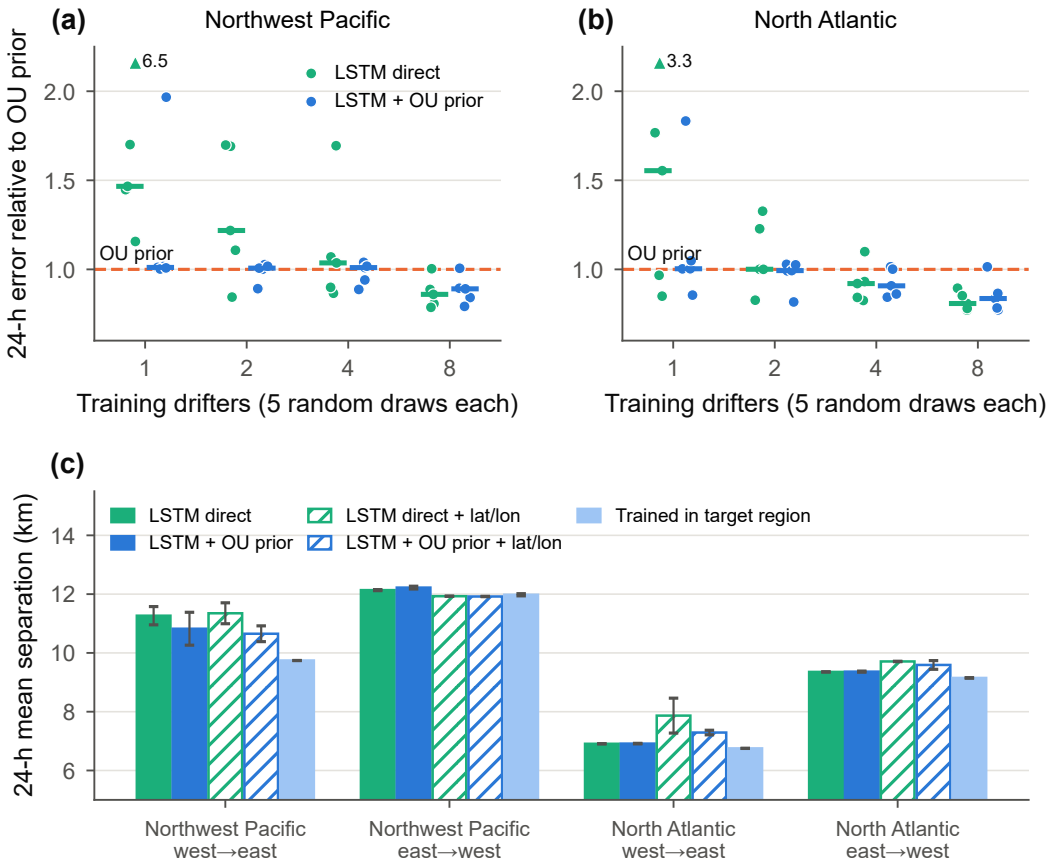


Figure 5. Role of the OU prior. (**a**,**b**) 24-h error of networks trained on 1, 2, 4 or 8 randomly drawn drifters (five draws each), relative to an OU prior fitted on the same drifters; bars show medians and triangles values beyond the axis. (**c**) 24-h mean separation for spatial transfer between the western and eastern sub-regions, without location inputs (filled), with location inputs (hatched) and for a prior-based model trained in the target sub-region (light blue); error bars show half the range over two seeds.

4.6. Calibrated Search Regions

Without calibration the Gaussian forecasts are miscalibrated: between 0.78 and 0.98 of the 72-h positions fall in the nominal 90% region of the OU prior, depending on basin and setting, while the LSTM ensemble covers 0.88 in the NWP and 0.91 in the NA, with group values between 0.79 and 0.93 (Table 4, Figure 6a,b). Static conformal calibration on 2015–2021 drifters does not fix this: in 2022 the LSTM coverage is 0.88 in the NWP and 0.92 in the NA in-basin, and 0.81–0.95 under transfer, because the calibration years are not exchangeable with the test year. The rolling variant restores marginal coverage to 0.89–0.91 in all settings, and the Mondrian variant additionally equalises coverage across groups: in-basin and for west-to-east transfer every latitude–drogue group with at least 200 test windows lies between 0.889 and 0.909 in both basins. For east-to-west transfer the worst group reaches 0.88 in the NA, but only 0.79 in the NWP, where two target groups contain few test forecasts and the rolling windows are often too small.

At comparable coverage (0.89–0.90), the hybrid model needs a median 72-h search region of 16.5×10^3 km^2 in the NWP and 6.8×10^3 km^2 in the NA, against 27.8×10^3 and 12.5×10^3 km^2 for the OU prior (41% and 46% smaller); the reductions are 29–44% for the transfer settings (Figure 6c) and 47–61% at 24 h, where both models again reach an empirical coverage of 0.90 (Table 5). Figure 7 shows two examples in which the LSTM follows a turning trajectory that the prior extrapolates almost linearly.

Table 4. Coverage of the 90% region at 72 h on the 2022 test windows (overall, and range over the latitude–drogue groups with at least 200 test windows) and median region area at 72 h (10^3 km^2) for different calibration methods.

Setting	Calibration	LSTM + OU prior			OU prior	
		Coverage	Group range	Area	Coverage	Area
Northwest Pacific						
In-basin	None	0.88	0.79–0.93	14.1	0.87	23.4
	Static	0.88	0.80–0.93	14.4	0.93	33.8
	Rolling	0.90	0.82–0.94	16.2	0.90	26.5
	Rolling Mondrian	0.90	0.89–0.91	16.5	0.90	27.8
West → East	None	0.93	0.90–0.97	16.3	0.93	28.7
	Static	0.94	0.92–0.97	17.8	0.97	44.3
	Rolling	0.90	0.87–0.95	14.2	0.90	23.3
	Rolling Mondrian	0.90	0.89–0.91	14.1	0.90	23.5
East → West	None	0.83	0.52–0.86	15.8	0.78	20.8
	Static	0.81	0.50–0.85	14.6	0.80	22.2
	Rolling	0.90	0.64–0.94	22.6	0.91	40.1
	Rolling Mondrian	0.89	0.79–0.92	21.6	0.89	30.8
North Atlantic						
In-basin	None	0.91	0.89–0.92	7.3	0.96	25.1
	Static	0.92	0.90–0.93	7.6	0.95	21.2
	Rolling	0.90	0.88–0.91	6.9	0.90	12.5
	Rolling Mondrian	0.90	0.90–0.90	6.8	0.90	12.5
West → East	None	0.95	0.92–0.97	8.4	0.98	33.8
	Static	0.95	0.92–0.97	8.1	0.97	25.3
	Rolling	0.90	0.86–0.93	5.8	0.90	10.0
	Rolling Mondrian	0.90	0.90–0.90	5.8	0.90	10.3
East → West	None	0.85	0.75–0.90	8.5	0.81	10.9
	Static	0.85	0.76–0.91	8.8	0.84	12.8
	Rolling	0.90	0.82–0.94	11.1	0.90	18.0
	Rolling Mondrian	0.90	0.88–0.90	10.5	0.89	14.8

Table 5. Coverage of the 90% region at 24 h on the 2022 test windows after rolling Mondrian calibration, median region area at 24 h (km^2) and relative reduction of the area of the LSTM with OU prior with respect to the OU prior.

Setting	LSTM + OU prior		OU prior		Reduction
	Coverage	Area	Coverage	Area	
Northwest Pacific					
In-basin	0.90	915	0.90	2,291	60%
West → East	0.90	926	0.90	1,981	53%
East → West	0.90	1,242	0.90	2,878	57%
North Atlantic					
In-basin	0.90	459	0.90	1,180	61%
West → East	0.90	390	0.90	1,000	61%
East → West	0.90	714	0.90	1,341	47%

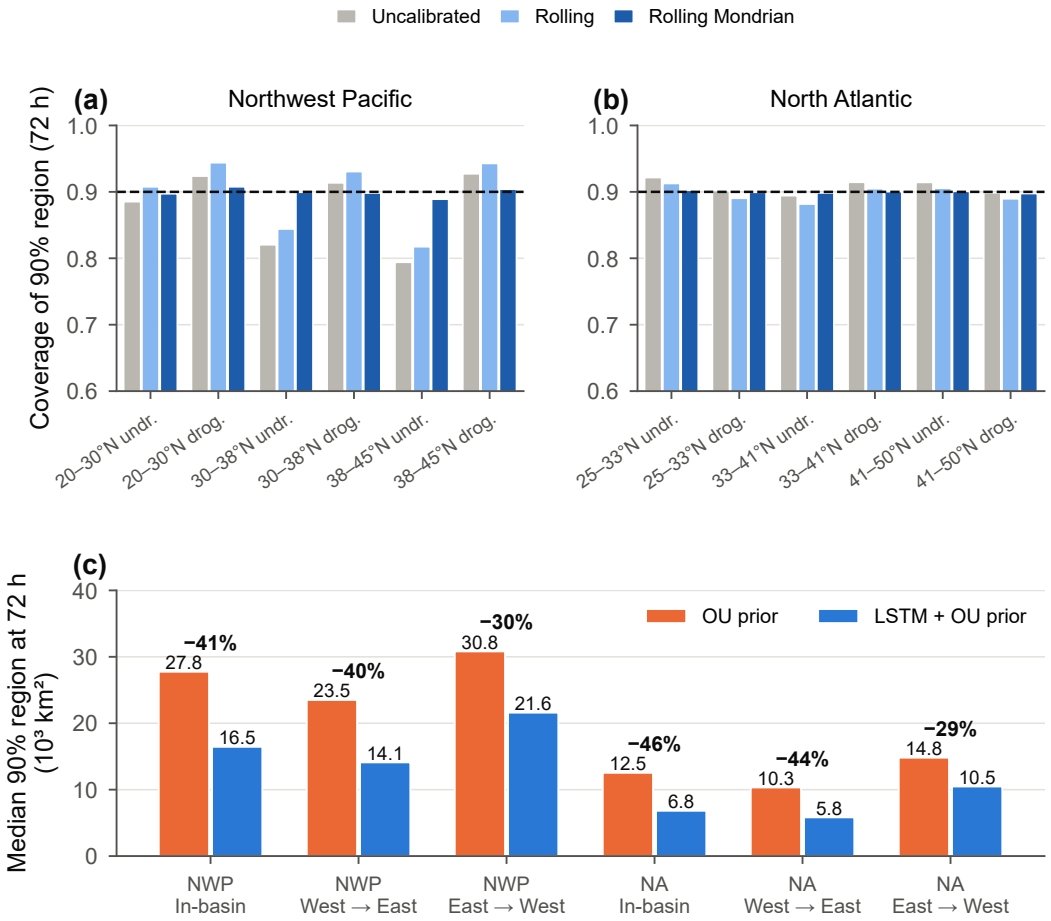


Figure 6. (**a**,**b**) Empirical coverage of the nominal 90% region at 72 h by latitude band and drogue state (undr.: undrogued; drog.: drogued) for the LSTM with OU prior on the 2022 test windows, before calibration, with rolling calibration and with rolling Mondrian calibration. (**c**) Median area of the 90% region at 72 h after rolling Mondrian calibration, at which both models reach an empirical coverage of 0.89–0.90, in-basin and for transfer between sub-regions (NWP: Northwest Pacific; NA: North Atlantic).

5. Discussion

5.1. Information Available in a Real Search

In a real search the object provides only the GPS fixes received up to the forecast time, and no product velocity is available. The comparison of input velocities shows that this distinction matters: in both basins, a velocity that uses positions up to 6 h after the forecast time reduces the error by an amount comparable to the gain of the deep network. When forecast models are developed from interpolated drifter products, recomputing velocities from past positions and separating training and test periods brings the evaluation closer to operational use, although the smoothed positions themselves may retain some look-ahead (Section 4.4). The same consideration applies to gridded current products that assimilate the drifters being predicted.

5.2. What the Network Learns, and Its Limits

The relative gain of the LSTM over damped velocity is nearly identical in the two basins (32% at 24 h), even though the absolute errors in the NA are about 27% lower. The network reduces errors most for drogued drifters, whose motion follows the 15-m currents more closely, whereas undrogued drifters are subject to larger direct wind forcing [30,36]. The location ablation suggests that it extracts persistence—curvature, recent changes of

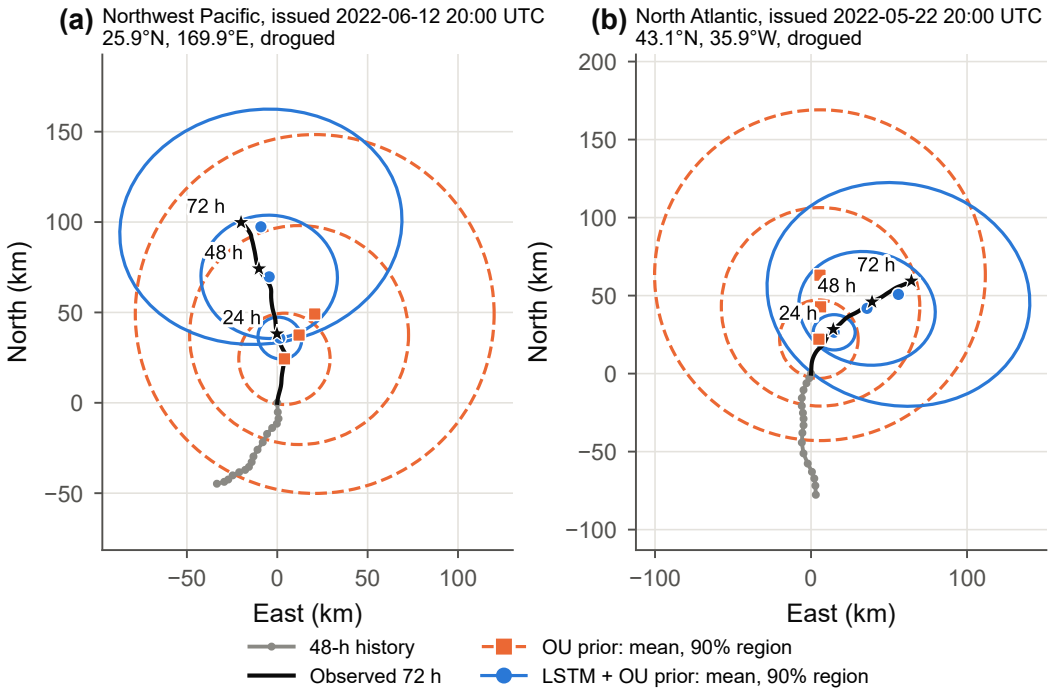


Figure 7. Two test forecasts in local coordinates centred on the drifter position at t_0. Grey: 48-h history; black: observed 72-h trajectory, with stars at 24, 48 and 72 h. Markers and ellipses: predicted means and calibrated 90% regions at 24, 48 and 72 h for the OU prior (orange, dashed) and the LSTM with OU prior (blue). The 72-h separations are 65.0 vs. 11.2 km (**a**) and 58.7 vs. 12.0 km (**b**).

speed and direction—from the history rather than memorising regional currents. No improvement was observed in this comparison when the training period was extended: training the same NWP model on 2020–2021 only (106,000 instead of 231,000 windows) gives a similar 24-h error (10.24 vs. 10.30 km), and other architectures do not improve on it either: an MLP and a sequence-to-sequence model with attention reach the same accuracy within 4% (Section 4.3). For the data volumes and architectures tested here, the skill extracted from the trajectory alone thus shows little further gain, and additional wind, wave or ocean-model information may be needed for further improvement [15].

5.3. Role of the Physical Prior

The OU prior encodes the simplest physics of drifter motion: a decorrelation time of 12–24 h and a partially persistent background flow, consistent with Lagrangian time scales reported for surface drifters [11]. It does not improve accuracy when data are plentiful, but in our experiments it made large errors much less frequent when only one or two drifters were available—precisely the situation of a forecast in a new region or for an unusual object—at the cost of slightly slower learning with moderate data. With one drifter, the prior-based network was more than 10% worse than the prior in 2 of 10 draws, against 8 of 10 for the direct network; with two drifters, in none against 6 of 10. It also provides the analytic reference against which the search-region reduction is measured.

5.4. Calibration under Distribution Shift

Interannual changes in eddy activity and drifter populations are enough to break the exchangeability assumption of split conformal prediction. Rolling calibration on recently verified forecasts is simple to operate, because GDP positions are distributed in near real time as raw fixes; the smoothed hourly product used here, in contrast, is released in periodic versions (version 2.01, with data up to October 2022, in September 2023), so an operational system would verify forecasts against the raw fixes. The Mondrian refinement reduces the

risk that the overall coverage hides under-coverage in energetic regions or for undrogued drifters, which are the most relevant proxies for debris. Its limitation is the number of verified forecasts per group; finer fall-backs, such as merging neighbouring groups, or adaptive conformal inference [34] may help in sparse regions.

5.5. Limitations

This study covers two ocean basins and a single test year, which limits the assessment of broader temporal and geographical generalisation. The forecasts rely exclusively on trajectory history without wind, wave or ocean-model forcing. Therefore, the results characterise the predictability available from drifter trajectories rather than the performance of a complete operational search-and-rescue system. In addition, the inputs are positions from a smoothed, periodically released product: velocities were recomputed from past positions, but the positions themselves, the verifying positions and the drogue state (from reanalysed drogue-loss dates [30]) may contain information that is not available at the issue time, so the evaluation is retrospective; a test with positions reconstructed from the fixes received before t_0 remains to be done.

6. Conclusions

This study developed a probabilistic drift forecasting framework based on historical drifter trajectories, combining an Ornstein–Uhlenbeck (OU) prior, an LSTM-based Gaussian correction and rolling Mondrian conformal calibration. The framework was evaluated for 6–72 h forecasts using GPS-tracked drifter trajectories from the Northwest Pacific and North Atlantic.

On the 2022 test sets, the hybrid model reduced the mean position error of damped-velocity extrapolation by 32% at 24 h and 18–19% at 72 h in both basins. Alternative neural architectures achieved comparable accuracy, indicating that the improvements were not specific to the LSTM architecture. The OU prior provided little additional accuracy with abundant training data but substantially improved robustness when only one or two drifters were available for training.

Rolling Mondrian conformal calibration brought the overall empirical coverage close to the nominal 90% level and improved coverage consistency across latitude–drogue groups. At comparable coverage, the hybrid model reduced the median search-region area by 47–61% at 24 h and 29–46% at 72 h relative to the OU prior. These findings demonstrate the potential of combining trajectory-based learning with adaptive uncertainty calibration for probabilistic drift forecasting and maritime search-area estimation.

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Abbreviations

The following abbreviations are used in this manuscript:

GDP	Global Drifter Program
GPS	Global Positioning System
LOWESS	Locally weighted scatterplot smoothing
LSTM	Long short-term memory
MLP	Multilayer perceptron
NA	North Atlantic
NLL	Negative log-likelihood
NWP	Northwest Pacific
OU	Ornstein–Uhlenbeck
RMS	Root-mean-square
SAR	Search and rescue

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